

Lecture 7

Sampling Distribution and Confidence Intervals

Part 2

CONFIDENCE INTERVALS

June 2026

- ◆ Consider the example in previous video, where the mean of the Netflix viewing hours was 6 hours, the standard deviation was 1,2 hours, and we take samples of size $N=100$.
- ◆ We know that the best estimate for the average of the sample mean is 6 hours, but it is very likely that for a particular sample, the value will be different.
- ◆ Intuitively the question is: if I take a simple sample, in which range of values I can expect to find my sample average?

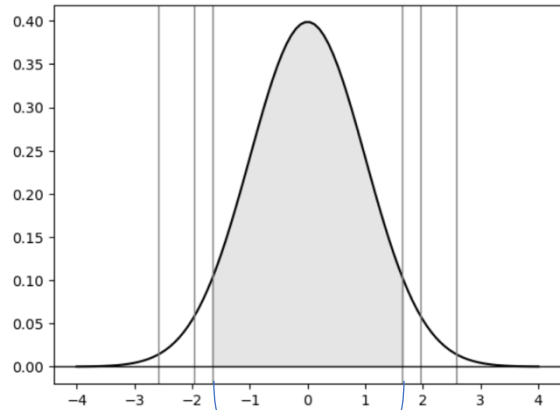
That question is not very well posed.

We will prefer to ask the following:

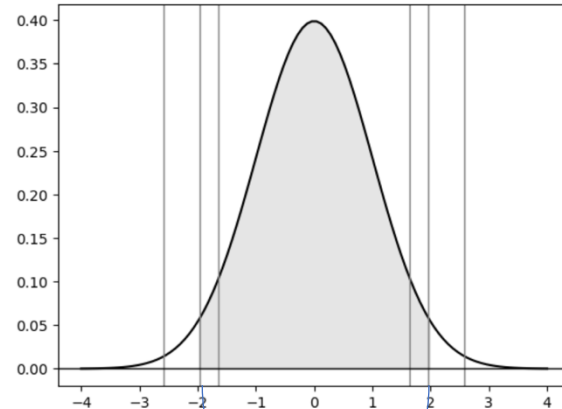
- ◆ What is a 95% confidence interval in which I will find my sample mean?
- ◆ Which can be rephrased as follows: Suppose that I draw many random samples of size N , each of them giving me a different sample average. I would like to select an interval of values such that 95 % of the times, the measured sample average is within that interval.
- ◆ Smaller interval $\Rightarrow$ less chance to be there $\Rightarrow$ less confidence (but more precision)
- ◆ Biggest interval $\Rightarrow$ more chance to be there $\Rightarrow$ more confidence (but less precision)

Important fact to remember. The distribution of samples means (when N is large) is approximately a normal distribution.

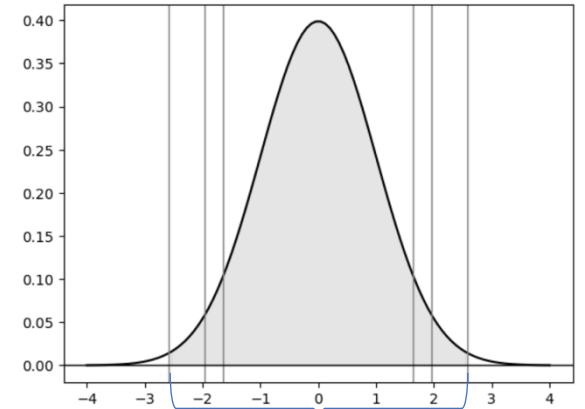
Confidence Intervals of the normal distribution



90% of data between
-1.645 and 1.645 std



95% of data between
-1.96 and 1.96 std



99% of data between
-2.58 and 2.58 std

Confidence Intervals of the standard normal distribution

In python is very easy to find any interval

```
from scipy import stats
ci_75=stats.norm.interval(0.75)
ci_80=stats.norm.interval(0.8)
ci_90=stats.norm.interval(0.9)
ci_95=stats.norm.interval(0.95)
print('75% interval: ', ci_75)
print('80% interval: ', ci_80)
print('90% interval: ', ci_90)
print('95% interval: ', ci_95)
```

```
75% interval:  (-1.1503493803760079, 1.1503493803760079)
80% interval:  (-1.2815515655446004, 1.2815515655446004)
90% interval:  (-1.6448536269514729, 1.6448536269514722)
95% interval:  (-1.959963984540054, 1.959963984540054)
```

Confidence Intervals for non standard normal distributions

Suppose that I am sampling data from a distribution with mean=30 and standard deviation=5, and the size of each sample is N=50

What is a 95% confidence interval for the sample mean?

Step 1: Find the value for the 95% ci of the standard normal distribution => 1.96

Step 2: Compute the standard error $se = \frac{\sigma}{\sqrt{N}} = \frac{5}{\sqrt{50}} = 0.7$

Step 3: Create an interval centered around the mean using the previous values

$$CI = [30 - 1.96 \times 0.7, 30 + 1.96 \times 0.7] = [\mathbf{28.628}, \mathbf{31.372}]$$

Example 1

♦ In the Netflix example, what is a 95% CI or the sample mean?

1. Take the population mean $\rightarrow \mu = 6$ hours
2. Compute the standard error, $SE = \frac{1,2}{\sqrt{100}} = 0,12$ hours
3. Select the value corresponding to 95% $\rightarrow 1,96$
4. Create the interval

$$CI = (\mu - 1,96 SE, \mu + 1,96 SE) = (6 - 0,2352, 6 + 0,2352) = \mathbf{(5,7648, 6,2352)}$$

95% if the samples will have a sample mean between 5,7648 and 6,2352

These values are so common that you should remember them

$$90\% \text{ confidence interval: } \bar{x} \pm 1.645 \frac{\sigma}{\sqrt{n}}$$

$$95\% \text{ confidence interval: } \bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

$$99\% \text{ confidence interval: } \bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$$

For any other confidence level, we can use Python to find the appropriate cut-off